

# 1. Accelerating Frame and Rindler Coordinates

$$\begin{aligned}t &= \left(\frac{c}{g} + \frac{x'}{c}\right) \sinh\left(\frac{gt'}{c}\right) \\x &= c\left(\frac{c}{g} + \frac{x'}{c}\right) \cosh\left(\frac{gt'}{c}\right) - \frac{c^2}{g} \\y &= y' \\z &= z',\end{aligned}$$

a) Let's start by computing the differentials. Let  $\rho \equiv \frac{c}{g} + \frac{x'}{c}$  for brevity, so:

$$\begin{aligned}dt &= \frac{g}{c}\rho \cosh\left(\frac{gt'}{c}\right) dt' + \frac{1}{c}\sinh\left(\frac{gt'}{c}\right) dx' \\dx &= g\rho \sinh\left(\frac{gt'}{c}\right) dt' + \cosh\left(\frac{gt'}{c}\right) dx' \\dy &= dy', \quad dz = dz'\end{aligned}$$

The Minkowski line element is  $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$ . This quantity is invariant hence we can try substituting:

$$\begin{aligned}ds^2 &= -c^2 \left( \frac{g^2}{c^2} \rho^2 \cosh^2\left(\frac{gt'}{c}\right) dt'^2 + 2\frac{g}{c^2} \rho \sinh\left(\frac{gt'}{c}\right) \cosh\left(\frac{gt'}{c}\right) dx' dt' + \frac{1}{c^2} \sinh^2\left(\frac{gt'}{c}\right) dx'^2 \right) \\&+ \left( g^2 \rho^2 \sinh^2\left(\frac{gt'}{c}\right) dt'^2 + 2g\rho \sinh\left(\frac{gt'}{c}\right) \cosh\left(\frac{gt'}{c}\right) dx' dt' + \cosh^2\left(\frac{gt'}{c}\right) dx'^2 \right) + dy'^2 + dz'^2\end{aligned}$$

Using  $\cosh^2 - \sinh^2 = 1$ ,

$$\begin{aligned}ds^2 &= -g^2 \rho^2 dt'^2 + dx'^2 + dy'^2 + dz'^2 \\ds^2 &= -\left(1 + \frac{gx'}{c^2}\right)^2 c^2 dt'^2 + dx'^2 + dy'^2 + dz'^2\end{aligned}$$

b) Using  $\sinh\left(\frac{gt'}{c}\right) \approx \frac{gt'}{c}$  and  $\cosh\left(\frac{gt'}{c}\right) \approx 1$ . We expand  $x$  to second order in  $t'$ :

$$x \approx \left(\frac{c^2}{g} + x'\right) \left(1 + \frac{1}{2} \frac{g^2 t'^2}{c^2}\right) - \frac{c^2}{g} = x' + \frac{1}{2} g t'^2 \left(1 + \frac{gx'}{c^2}\right)$$

To leading (Newtonian) order:  $x \approx x' + \frac{1}{2} g t'^2$ , i.e. uniform acceleration  $\ddot{x} = g$  from rest.

c) For a clock at rest at position  $x'$  in the prime frame,  $dx' = dy' = dz' = 0$ , so proper time is:

$$d\tau = \sqrt{-\frac{ds^2}{c^2}} = \left(1 + \frac{gx'}{c^2}\right) dt'$$

Thus:

- Clock at  $x' = 0$ :  $d\tau_0 = dt'$
- Clock at  $x' = h$ :  $d\tau_h = \left(1 + \frac{gh}{c^2}\right) dt'$

The ratio is:

$$\frac{d\tau_h}{d\tau_0} = 1 + \frac{gh}{c^2}$$

d) At a fixed  $t'$ , the spatial metric is  $d'^2 = dx'^2 + dy'^2 + dz'^2$ . So the distance between  $x' = 0$  and  $x' = h$  is  $\int_0^h dx' = h$ , which is a constant, so its rigid motion as its independent from time.

e) From part (c), proper time at  $x'$  runs at rate  $\left(1 + \frac{gx'}{c^2}\right)$  relative to coordinate time  $t'$ . The observer at  $x' = 0$  is defined to have proper acceleration  $g$ . Any observer at  $x'$  has the same coordinate motion scaled by the factor  $\left(1 + \frac{gx'}{c^2}\right)$ , visible from the transformation since  $\rho = \frac{c}{g} + \frac{x'}{c}$  just rescales everything. Proper acceleration then scales inversely, giving

$$\frac{g}{1 + \frac{gx'}{c^2}}$$

## 2. Hafele-Keating Experiment

a) Work in an inertial frame centred on Earth's centre. A surface clock at radius  $R_\oplus$  moves with speed  $V_\oplus$  due to Earth's rotation. A flying clock is at altitude  $h(t)$  and has ground speed  $V_g(t)$  relative to the surface, so its speed in the inertial frame is  $V_g + V_\oplus$  (for eastward flight).

The metric outside Earth (weak field, slow motion) gives proper time:

$$d\tau \approx \left(1 + \frac{\phi}{c^2} - \frac{v^2}{2c^2}\right) dt$$

where  $\phi = -\frac{GM}{r}$  is the gravitational potential and  $v$  is the speed in the inertial frame. Taking  $\phi \approx -\frac{GM}{R_\oplus} + gh$  with  $g = \frac{GM}{R_\oplus^2}$

Flying clock:

$$d\tau_{fly} \approx \left(1 - \frac{GM}{R_\oplus c^2} + \frac{gh}{c^2} - \frac{(V_g + V_\oplus)^2}{2c^2}\right) dt$$

Surface clock:

$$d\tau_{surf} \approx \left(1 - \frac{GM}{R_{\oplus}c^2} - \frac{V_{\oplus}^2}{2c^2}\right) dt$$

The difference  $\Delta\tau = \int (d\tau_{fly} - d\tau_{surf})$ :

$$\Delta\tau = \frac{1}{c^2} \int dt \left[ gh(t) - \frac{1}{2} V_g(t)(V_g(t) + 2V_{\oplus}) \right]$$

b) Use typical values:  $h \approx 10km$ ,  $V_g \approx 800 \frac{km}{h} \approx 222 \frac{m}{s}$ ,  $g \approx 9.8 \frac{m}{s^2}$ ,  $V_{\oplus} \approx 465 \frac{m}{s}$ , and a circumnavigation takes  $T \approx \frac{40000}{800} \approx 50hr = 1.8 \times 10^5 s$ , with 40000 the circumference of the earth.

Gravitational term (same for both directions):

$$\frac{ghT}{c^2} \approx +204ns$$

Kinematic term:

$$-\frac{T}{2c^2} V_g(V_g + 2V_{\oplus})$$

- Eastward  $V_g = +222$ ,  $V_{\oplus} : \approx -270ns$
- Westward  $V_g = -222$ ,  $V_{\oplus} : \approx +166ns$

Then we have a total  $\Delta\tau$  of -66 ns for eastward, and 370 ns for west ward.